

Project Varuna / FloodTwin — Novelty & Prior-Art Search

Author: Ansh Vivek (independent researcher) · anshvivek2003@gmail.com **Date:** 23 July 2026 **System under study:** *Varuna FloodTwin* — an open, free-tier, satellite-built differentiable urban flood digital twin with street-graph drainage planning, a self-supervised street-graph GNN, buildability-constrained detention siting, managed-aquifer-recharge coupling, and a reward-gated citizen-report learning loop. Live: [project-varuna-iota.vercel.app](#) · API: [anshvivek-varuna-floodtwin.hf.space](#) · Code: [github.com/Maverick-Ansh/project-varuna](#)

0. How this search was done (and how honest it is)

This is a *prior-art / novelty* search, not a legal freedom-to-operate opinion. Five parallel research agents swept five sub-domains — (1) flood digital twins & differentiable hydraulic models, (2) AI/GNN flood prediction & flood-aware routing, (3) urban drainage optimisation & managed aquifer recharge, (4) SAR flood validation, DEM uncertainty, citizen science & night-lights, and (5) patents. Every URL below was returned by live web/patent search on 2026-07-23 and the load-bearing items were fetched and verified. Where a page was paywalled or auto-fetch was blocked (anti-bot), the entry is the canonical DOI/publisher landing page and is flagged. **No citations or patent numbers were fabricated.** Full raw links are in the [Sources appendix](#).

Honest caveats up front. (a) The *closest single work to Varuna's differentiable core* — “*Inunda*” (arXiv, July 2026) — is essentially concurrent, not something Varuna predates. (b) Several Varuna ingredients (the Bates-2010 solver, FILM, HAND/TWI indices, VIIRS Black Marble night-lights, ASR/MAR) are well-established and are cited here as *methods/baselines*, **not** claimed as novel. (c) Varuna's headline metrics (CSI $\approx$ 0.033, 80.9 % Patna cut, $\sim$ 5 % vs $\sim$ 52 % co-location) are its own results; no external work reports those exact diagnostics, which supports novelty but means there is no third-party corroboration of the numbers themselves.

1. What we (Project Varuna) have done — the novelty anchors

Nine contributions, each stated as a testable claim. Sections 2–3 test each against the field.

#	Varuna component	The specific claim to defend
V1	Free-tier satellite–twin–dashboard pipeline	An end-to-end differentiable 2-D local-inertial shallow-water flood twin (Bates 2010 scheme in PyTorch) built per-city from only free global data (FARDEM, ESA WorldCover, JRC water, Sentinel-1), distilled into a U-Net emulator for millisecond what-ifs, and served as a live public dashboard — the whole build+serve on free compute (Colab/Kaggle T4, CPU serve), $\sim$ 15–25 min/city.
V2	Honest SAR validation + DEM-uncertainty diagnosis	Against 8 Sentinel-1 masks the twin scores a negative CSI = -0.033 – -0.041 , statistically indistinguishable from TWI/HAND; a ± 1 m spatially-correlated DEM ensemble explains <i>why</i> — only $\sim$ 5 % of flooded area co-locates on flat deltaic terrain vs $\sim$ 52 % in hills. A terrain-data-regime diagnosis of when no topographic router can co-locate flat-city ponding.
V3	Street-graph “spiderweb” drainage, benefit re-simulated	Storm drains routed on the real OSM road graph via one reverse multi-source Dijkstra from flood-safe outfalls (never the river) , $\sim$ 40 inlets at basin lows, carved descending beds, then re-simulated to measure a 73–89 % street-flood cut at better cost/m ³ than sparse canals.
V4	FloodGNN — self-supervised street-graph GNN	Edge-conditioned message-passing GNN over the OSM street graph, FILM-conditioned on rainfall (one model, continuous storm range), trained only on labels the twin generates (zero flood ground-truth); per-street risk in a 4 ms pass ($\sim$ 60 $\times$ faster than raster), driving a flood-safe evacuation-routing endpoint, with zero-shot flat–hilly cross-city transfer (AUC 0.72–0.85).
V5	Buildability-constrained phased detention siting	Detention containers sited only on physically-available ground (building footprints & permanent water excluded, under-street allowed), each sized to its local pooled depth, flood-cut re-measured , delivered as a 10/20/40 %–cut phased, costed ladder; top sites hit each city's notorious flood spots with zero tuning.
V6	Flood-to-drought / aquifer-recharge coupling	A single free-data pipeline that couples urban flood reduction with managed aquifer recharge — matching pooled water to a depleted aquifer via a Recharge Suitability Index — framed to India's Atal Bhujal Yojana & CPHEEO norms; V2 “Bhujal” adds metered recharge plans for Karnataka towns.
V7	Reward-gated citizen-report online learning	Rate-limited citizen depth-band reports $\rightarrow$ nightly emulator fine-tuning behind an auditable reward gate (band-tolerant loss + simulation-replay anchor against the all-wet failure + held-out-by-day $\geq$ 5 % improvement + $\geq$ 70 % bootstrap win-rate); every accept/hold decision written to a public learning log.
V8	Reproducibility-hazard finding (seeded collapse)	Fixed-seed data generation silently made network init bit-identical, deterministically collapsing the softmax emulator head to “never flood” on a half-ocean tile (validation RMSE frozen to the last digit); a detect-and-reseed guard turned the worst tile into the best emulator. A documented “ fixed seeds make failures reproducible too ” result with a frozen-metric test.
V9	Grounded multilingual advisories + VIIRS outage overlay	English/Hindi/Marathi advisories generated from only the system's own numbers (strict-JSON, deterministic fallback), plus a VIIRS Black Marble night-radiance-drop overlay flagging candidate power outages (with a “unknown $\neq$ dark” monsoon cloud-mask caveat).

2. What *others* have done — prior art by theme

Each table: **Work** (hyperlinked title, authors, year, venue) · **What it does** · **Relation to Varuna & the key difference**. "≈" in the verdict column of §3 quantifies closeness.

2.1 Flood digital twins & differentiable / physics-based flood models

Work	What it does	Relation & key difference from Varuna
Inunda: a GPU-native differentiable flood solver — Zhi Li, 2026, arXiv preprint	Bates-2010 local-inertial 2-D solver fully in PyTorch ; calibrates Manning's n / bed / infiltration by autodiff-through-the-time-loop + gradient descent.	Closest single work — nearly Varuna's exact core (Bates + PyTorch + differentiable roughness). Differs: US datasets, <i>pure solver</i> — no U-Net emulator, no hosted twin/dashboard, no free-global-data developing-world framing. Near-concurrent (July 2026): related work, not predated.
Universal Shallow Water Equations with Differentiable Programming — Liu et al., 2025, Water Resources Research	Embeds a neural flow-resistance closure inside the 2-D SWE, trained end-to-end via differentiable programming.	Very close on differentiable roughness. Differs: full SWE (not reduced local-inertial), a methods paper on a closure — no emulator, twin, or open-data city pipeline. (<i>Paywalled; DCI verified.</i>)
A simple inertial formulation of the shallow water equations — Bates, Horritt & Fewtrell, 2010, J. Hydrology	The foundational local-inertial (LISFLOOD-ACC) scheme Varuna implements.	This is Varuna's numerical core. Differs: original is non-differentiable Fortran/C. Varuna's novelty is the PyTorch, gradient-optimisable re-implementation — not the scheme itself (cited as method).
LISFLOOD-FP 8.0 — Shaw, Sharifian, Bates, Neal et al., 2021, GMD	Operational descendant of Bates-2010: DG/acceleration solvers, GPU, subgrid channels.	Direct lineage & gold-standard numerical baseline. Differs: classic HPC code — no autodiff, ML emulator, or web twin.
Subgrid local-inertial formulation for urban flood simulation — Nithila Devi & Kuiry, 2024, WRR	Improves local-inertial equations with subgrid topography for urban flooding.	Same equation family, same urban target. Differs: numerical-methods advance only — not differentiable, no emulator/twin.
SFINCS — Deltaires (Leijnse et al.), 2020–	Reduced-complexity compound-flood solver; minutes-fast; widely operational.	Closest open-source reduced-complexity engine in spirit. Differs: Fortran, non-differentiable, no gradient calibration, no U-Net emulator; a solver library, not a hosted free-data city twin.
Fathom Global Flood Map — Fathom (Bristol), 2013–	Global 30 m flood-hazard maps on a LISFLOOD-FP-derived engine + ML; industry benchmark.	Closest commercial global product, shares Bates lineage. Differs: proprietary, paid, static hazard layers — not open, differentiable, interactive, or free-tier reproducible.
JBA Global Flood Models — JBA Risk Mgmt	Global catastrophe/flood hazard maps & analytics for insurance/government.	Commercial peer. Differs: proprietary risk product — no differentiable physics, open-data twin, or emulator dashboard.
Previsico Flood Intel Platform — Previsico	Real-time property-level surface-water flood forecasting on a bespoke hydrodynamic suite.	Closest commercial real-time flood twin. Differs: closed, subscription, developed-world markets; no differentiable/open-data stack, no free-tier serving.
FloodCast / GeoPINS — Xu, Zhu, Bamber et al., 2024, Water Research	Couples multi-satellite obs with a geometry-adaptive physics-informed neural operator to solve SWE fast at large scale (Pakistan 2022).	Very close: physics + ML + satellite for rapid flood modelling. Differs: PINN/neural-operator PDE solver (not differentiable finite-difference + separate U-Net), basin-scale not per-city open twin dashboard.
Deep CNN for rapid fluvial flood inundation — Kabir, Patidar, Xia, Liang et al., 2020, J. Hydrology	CNN trained on hydrodynamic outputs to predict depth maps in near-real-time.	Precedent for Varuna's U-Net emulator. Differs: emulator only, teacher solver non-differentiable, no open-global-data pipeline/twin.
DassFlow2D-V3: variational DA with algorithmic-differentiation adjoint — Pujol, Monnier, Lanier et al., 2022, GMD	2-D SWE finite-volume model whose adjoint (auto-differentiated) assimilates flood/satellite data.	Closest <i>differentiable full-hydraulic</i> lineage predating Varuna. Differs: adjoint of Fortran (not native PyTorch autodiff), full SWE, research assimilation tool — no ML emulator or public twin.
Adjoint-based assimilation for 2-D urban flood parameters — Zhang, Monnier, Garambois et al., 2024, J. Hydrology	Adjoint/differentiable 2-D urban hydraulic model inferring distributed roughness via sensitivity maps.	Very close on differentiable-hydraulics-for-urban-parameter-inference. Differs: adjoint-code, no emulator, no open-global-data free-tier twin.
smash v1.0: differentiable regionalizable hydrological modelling — Colleoni, Garambois, Jay-Allemand et al., 2025, GMD	Differentiable distributed hydrology + NN-regionalised parameters + variational DA.	Same differentiable-programming paradigm. Differs: rainfall-runoff hydrology, not 2-D hydraulic inundation; no urban twin/emulator/dashboard.
Fully differentiable distributed rainfall-runoff modelling — 2025, HESS	End-to-end differentiable distributed rainfall-runoff model in an autodiff framework.	Anchors the “differentiable hydrology” movement motivating Varuna. Differs: hydrologic (runoff), not hydraulic inundation; no twin/emulator layer.
CLDNet: conditional latent-dynamics AI digital twin for metropolitan floods — 2026, arXiv preprint	Observation-coupled digital twin: fast low-dim latent SWE surrogate with meshless decoder queried at gauges.	Closest “AI digital twin + SWE surrogate” concept to Varuna's twin+emulator pair. Differs: latent-dynamics surrogate (not differentiable Bates + U-Net), no open-satellite developing-city framing.
FlowsDT: geospatial digital twin for urban flood dynamics — 2026, Computers, Environment & Urban Systems	Geospatial urban-flood digital twin integrating geodata + hydrodynamics for scenario navigation.	Direct “urban flood digital twin” peer. Differs: not differentiable, no PyTorch emulator, free-tier/open-global-data is not its thesis.
Digital Twins for Urban Flood Risk Management: a systematic review — 2025, Remote Sensing	Survey of digital-twin + remote-sensing + early-warning systems for urban floods.	Landscape/positioning citation; a review, not a competitor — useful to frame the open/differentiable/free-tier gap Varuna fills.
Probabilistic flood-risk mapping for Indian megacities (ALERT, GEE) — 2025	Scalable ML flood-hazard classification on free Google Earth Engine data across Chennai/Mumbai/Bengaluru/Kolkata/Gurugram.	Closest free-open-data-for-Indian-cities peer. Differs: ML hazard classification, not a differentiable hydraulic twin with drainage/recharge planning.
Copernicus/Sentinel-1 Global Flood Monitoring (GFM) — 2025, RSE	Fully automatic global S-1 SAR flood observation service.	Parallels Varuna's Sentinel-1 <i>input</i> , not its simulator. Differs: observation product, not a physics twin.

2.2 AI / GNN flood prediction, emulation & flood-aware routing (FloodGNN's field)

Work	What it does	Relation & key difference from Varuna's FloodGNN
Global prediction of extreme floods in ungauged watersheds — Nearing/Cohen et al. (Google Flood Hub), 2024, <i>Nature</i>	LSTM river-discharge model forecasting extreme streamflow globally incl. ungauged basins, 5-day lead.	Flagship AI-flood-forecasting citation. Differs: predicts <i>river-gauge hydrographs</i> per basin, not per-street urban inundation; no city graph, no routing, no simulator self-supervision.
Predictions in Ungauged Basins with Machine Learning — Kratzert, Klotz, Shalev et al., 2019, WRR	Multi-basin LSTM rainfall-runoff generalising to basins with no training data via catchment attributes.	Conceptual ancestor of zero-shot transfer. Differs: 1-D streamflow, not spatial street-graph inundation; gauge-supervised not twin-labelled; no GNN/routing.
SWE-GNN: hydraulics-based GNN flood modelling — Bentivoglio, Isufi, Jonkman, Taormina, 2023, HESS	GNN surrogate of the shallow-water equations on a finite-volume mesh graph, ~100× faster, generalises to unseen topographies.	One of the closest works. Differs: graph = hydrodynamic <i>mesh cells</i> not OSM roads; autoregressive depth rollout, not single-pass per-street risk; boundary-driven not FiLM-rainfall; no routing.
mSWE-GNN: multi-scale hydraulic GNN — Bentivoglio et al., 2025, NHESS	Multi-resolution successor generalising across meshes/topographies/BCs at 700–1000× speed-up.	Closest analog to Varuna's cross-topography generalisation. Differs: mesh-node autoregressive surrogate needing a fine-tuning sim per new site (vs zero-shot flat→hilly); no street graph, no routing.
FloodGNN-GRU: spatio-temporal GNN for flood prediction — Kazadi, Doss-Gollin, Sebastian, Silva, 2024, Environmental Data Science	GNN+GRU predicts depth/velocity over a mesh graph, trained/tested on a single Harvey sim, ~1000× faster.	Nearest namesake. Differs: temporal autoregressive on <i>one</i> storm (authors flag poor rainfall generalisation); mesh not roads; no FiLM continuous-storm model; no cross-city transfer; no routing.
Pluvial Flood Emulation with Hydraulics-informed Message Passing — Kazadi, Doss-Gollin, Da Silva, 2024, ICML	Hydraulics-informed message-passing GNN predicting depth autoregressively from terrain + precipitation.	Very close: simulation-trained rainfall-driven GNN emulator. Differs: full depth-field rollout on terrain graph, not per-street risk in one pass; no explicit FiLM; no routing/transfer.
ASTGCN: spatial-temporal graph DL for urban flood nowcasting — Farahmand, Xu, Mostafavi, 2023, Scientific Reports	Attention spatio-temporal GCN nowcasts census-tract inundation from rainfall + human reports (Harvey).	Close on GNN-over-urban-graph. Differs: supervised on <i>observed/reported</i> floods, not twin labels; nodes = census tracts not road segments; no continuous-rainfall single model; no routing/transfer.
Interpretable physics-informed GNNs for flood forecasting — Taghizadeh et al., 2025, CACIE	Physics-informed interpretable GNN for flood forecasting under hydraulic constraints.	Same family. Differs: forecasting, not street-graph risk + routing; no FiLM single model; no zero-shot city transfer.
GNN surrogate for real-time hydraulics of urban drainage networks — Zhang, Tian, Lu, Liao, Yuan, 2024, Water Research	Physics-guided GNN surrogate of a drainage flow-routing model predicting junction/conduit hydraulics.	Closest "GNN surrogate of a hydraulic sim on a real network graph." Differs: sewer/pipe graph not OSM streets; predicts hydraulics for control not per-street risk for evacuation; no routing/transfer/FiLM.
GNNs for route recommendation under probabilistic hazards — Liu & Meidani, 2025, arXiv	GNN approximates shortest paths/distances on transport networks under hazard disruptions (hurricane evacuation).	Closest work on the flood-aware routing axis. Differs: learns routing itself under generic hazard probabilities; no flood-risk-from-a-twin, no FiLM, no inundation transfer. Varuna feeds GNN risk into risk-inflated Dijkstra.
Flood numerical model + Dijkstra risk-avoidance routing — 2023, Water Resources Management	Couples a 2-D hydrodynamic model with road-segment weighting + classic Dijkstra for risk-avoidance routes.	Direct analog to Varuna's risk-inflated Dijkstra. Differs: risk from a <i>slow full sim per scenario</i> , not a learned one-pass street-graph GNN; no ML risk model, no transfer.
Emergency evacuation routing for dam-failure flooding — 2025, Applied Sciences	Couples flood evolution with a road-network model to optimise dam-failure evacuation routes.	Flood-aware routing prior art. Differs: simulation-coupled optimisation, no learned GNN risk predictor, no FiLM, no cross-city transfer.
Transferability of data-driven urban pluvial flood-depth models (Berlin) — Selem et al., 2023, NHESS	Tests spatial transfer of CNN/RF urban pluvial depth models to unseen city areas.	Directly on cross-city transfer. Differs: raster CNN/RF, not a street-graph GNN; transfer within one city, not flat→hilly zero-shot; no FiLM/routing.
Cross-region streamflow/flood forecasting at global scale (ED-DLSTM) — Li et al., 2024, The Innovation	Encoder-decoder double-LSTM for cross-region/ungauged streamflow + flood forecasts.	Supports spatial-transfer thread. Differs: 1-D streamflow transfer, not inundation transfer across street graphs; no GNN/routing/FiLM.
Data-driven flood emulation with deep CNNs — Guo, Leitão, Simões, Moosavi, 2021, J. Flood Risk Mgmt	CNN emulator of a hydrodynamic urban-flood model predicting max depth from rainfall + terrain rasters.	Anchor for Varuna's U-Net emulator. Differs: grid-CNN on rasters (not graph), single scenario; no street-graph risk/routing/FiLM/transfer.
U-RNN: high-resolution spatiotemporal urban-flood nowcasting — Jiang, Chen et al., 2025, J. Hydrology	Recurrent U-Net for dynamic spatiotemporal urban flood nowcasting.	Advanced surrogate in Varuna's emulator space. Differs: pure DL surrogate; no differentiable physics core, gradient calibration, or open-data twin.
Multi-fidelity GNNs for flood-hazard mapping — Bentivoglio group, 2025, Reliability Eng. & System Safety	Fuses low-/high-fidelity data in a GNN for efficient accurate hazard maps.	Recent GNN-flood-mapping alternative. Differs: fidelity fusion on mesh, not FiLM-rainfall street-risk + routing + zero-shot city transfer.
Deep learning for flood mapping: a review — Bentivoglio, Isufi, Jonkman, Taormina, 2022, HESS	Systematic review (58 papers) of DL for flood mapping incl. CNN/GNN surrogates.	Positioning/survey citation — supports "no prior system combines street-graph GNN + FiLM + twin self-supervision + routing + transfer."
FiLM: Feature-wise Linear Modulation — Perez, Strub, de Vries, Dumoulin, Courville, 2018, AAAI	Introduces affine (γ, β) conditioning of network features on side information.	The exact conditioning mechanism Varuna uses. Cited as method — novelty is <i>applying FiLM-on-rainfall to a street-graph flood GNN so one model spans a continuous storm range</i> .

2.3 Urban drainage-network optimisation & managed aquifer recharge (V3, V5, V6)

Work	What it does	Relation & key difference from Varuna
Graph-theory combinatorial MOO for (de)centralized drainage layouts — Hesarkazzazi et al., 2022, Sustainable Cities & Society	Uses the street network as base graph , then graph-theory + multi-objective optimisation to generate optimal sewer layouts.	Closest on <i>street-graph-based layout</i> . Differs: optimises pipe topology/cost against a hydraulic model; does not re-simulate a 2-D flood twin to <i>measure</i> street-flood-volume cut, no recharge coupling.
Structural optimisation of urban drainage systems — 2024, Water (MDPI)	Optimises drainage-network <i>structure</i> (looped/branched) to reduce flood risk.	Similar aim. Differs: structural/graph metrics, not OSM-street spiderweb with re-simulated benefit; no recharge.
Network structure optimisation considering pipeline redundancies — 2022, Int. J. Disaster Risk Science	Optimises redundancy/resilience of drainage layout.	Overlaps on resilience. Differs: resilience-metric focus, no flood-safe-outfall logic, detention siting, or recharge.
Hydraulic-based optimisation of stormwater drainage networks — 2024, Applied Water Science	Hydraulic optimisation of pipe diameters/slopes.	Overlaps on hydraulic sizing. Differs: sizes an assumed layout; Varuna <i>generates</i> layout on the real street graph & measures flood cut.
Hydrograph-based storm-sewer design optimisation by GA — Afshar et al., 2005, Can. J. Civil Eng.	GA coupled to SWMM to minimise storm-sewer construction cost.	Canonical GA+SWMM prior art. Differs: cost-min of pipe design, no street-graph routing, twin re-sim, or recharge.
Multi-objective optimisation of LID stormwater controls — 2018, J. Hydrology	Selection/placement/sizing of LID controls trading cost vs runoff.	Overlaps Varuna's detention siting. Differs: parcel-scale LID, no buildability/footprint-exclusion tied to pooled-depth sizing, no recharge index.
SWMMLIDopt: LID optimisation tool for SWMM — 2024, J. Hydroinformatics	Open tool to optimise LID selection/placement in SWMM for cost-effectiveness.	Same optimisation niche. Differs: LID sizing/cost only; no under-street tank siting by local pooled depth, phased ladders, or recharge.
Spatial optimisation of LID cost-effectiveness (SWMM-FTC) — 2022, J. Environmental Management	Spatial optimisation of LID siting for cost-effective runoff reduction.	Overlaps detention/storage siting. Differs: green-LID focus, no aquifer coupling, no re-simulated flood-volume measurement.
Review of Sponge City implementation in China — 2023, Water Science & Technology	Reviews China's Sponge City program performance & policy.	Umbrella concept behind flood-absorb-and-store. Differs: policy/LID-density review, not an algorithmic network-design + recharge-routing planner with measured per-city benefit.
Sponge City evidence from Shenzhen — 2025, Urban Science	Case evidence of sponge-city runoff/peak-flow reductions.	Real-world outcomes. Differs: empirical case study, not a transferable design engine; no flood-to-recharge water balance.
Enhanced Aquifer Recharge Using Stormwater: State of the Science — US EPA, 2021	Authoritative review of recharging aquifers with urban stormwater (methods, quality, clogging).	Underpins Varuna's flood-to-recharge idea. Differs: a science review, not a spatial planner routing pooled floodwater to a depleted aquifer via a Recharge Suitability Index.
MAR to alleviate urban floods + groundwater depletion — 2026, Science of the Total Environment	Uses MAR (well injection / ASTR) to <i>simultaneously</i> cut urban flood volume and raise water tables.	Closest MAR-coupling work — explicitly dual flood + groundwater. Differs: well-injection hydrogeology; Varuna adds street-graph drainage design + detention siting + re-simulated flood cut, framed to Indian norms.
Lessons Learned from ASR Systems in the US — GWPC, 2022	Synthesises ASR practice: store surplus/flood water, recover in drought.	Establishes ASR dual flood-drought precedent. Differs: engineered well ASR, not surface-pooling→recharge siting coupled to a city flood twin.
Innovative Drought and Flood Mitigation Projects — FEMA, 2017	Catalogs projects (incl. ASR) mitigating flood peaks while banking water for drought.	Same dual-hazard framing. Differs: policy/project catalog, not a computational design tool.
Atal Bhujal Yojana (Atal Jal) — CGWB / Min. of Jal Shakti, 2020–	India's flagship community groundwater-management + recharge scheme in 7 stressed states.	The policy Varuna is framed around. Differs: demand-side community governance, not an urban drainage-to-recharge engineering planner — Varuna operationalises recharge spatially .
Legislation on Rainwater Harvesting (Chennai/TN) — CSE India, 2003	Documents TN's mandatory building-scale RWH raising urban groundwater.	India urban recharge-mandate precedent. Differs: parcel-level mandated RWH, not network-scale floodwater routing to aquifer with measured flood reduction.
UTFI: taming floods for irrigation via underground storage (Ganges) — Pavelic et al. / IWMI, 2015	Introduces Underground Taming of Floods for Irrigation — divert high flows to recharge aquifers, cutting floods and supplying dry-season water.	Conceptually the single closest prior art to Varuna's flood-to-drought thesis . Differs: rural, river-basin scale via ponds/wells; Varuna is <i>urban</i> , works on the city street graph, engineers the drainage network, and re-simulates the measured street-flood cut.
Floodwater recharge to offset groundwater depletion (Ramganga) — 2017, Sustainable Water Resources Management	Couples SWAT+MODFLOW+HEC-RAS to test distributed MAR offsetting depletion while cutting flooding.	Same coupled flood+recharge ambition. Differs: basin-scale agricultural MAR, no urban drainage-network design or detention siting.
Managing underground transfer of floods for irrigation (Ramganga) — 2019, J. Hydrology	Field/modelling assessment of UTFI recharge performance & flood offset.	Empirical arm of flood-to-recharge. Differs: rural, pond-based; no urban street-graph engine or re-simulated urban flood cut.
Urban flood mitigation via deep aquifer recharge (Guadalajara) — Vanegas-Espinosa et al., 2022, IJERPH	Proposes routing <i>urban</i> floodwater into deep aquifer recharge to mitigate city flooding.	Closest urban flood→recharge coupling . Differs: conceptual/site-suitability for one city, no algorithmic street-graph drainage design, no re-simulated quantified flood cut, no detention cost-ladder.

2.4 SAR flood validation, DEM uncertainty, citizen science & night-lights (V2, V7, V9)

Work	What it does	Relation & key difference from Varuna
Validation of a 30 m flood-hazard model of CONUS — Wing, Bates, Sampson et al., 2017, WRR	Benchmarks a CONUS 2-D model vs observations; reports CSI up to ~0.87.	Canonical CSI validation. Differs: Varuna uses the same metrics but reports an <i>honest negative</i> on flat deltaic terrain and explains it, rather than a headline CSI.
First collective validation of global fluvial flood models (Nigeria, Mozambique) — Bernhofen et al., 2018, Environ. Research Letters	Inter-compares six global flood models vs satellite extent using CSI (~0.4–0.7).	Closest validation-of-many-models precedent. Differs: benchmarks but does not diagnose <i>why</i> CSI collapses; Varuna adds the DEM-ensemble co-location mechanism.
Sen1Floods11 — Bonafilia, Tellman, Anderson, Issenberg, 2020, CVPR-W	Global hand-labelled S-1 flood benchmark (4,831 chips, 11 events) for DL water segmentation.	Defines the S-1 water-mask task Varuna validates against. Differs: static supervised benchmark; Varuna validates a <i>dynamic physics twin</i> , using SAR as reference not training target.
Robustness of Bayesian S-1 flood mapping: multi-event validation — 2025, Science of Remote Sensing	Multi-event S-1 flood-mapping validation with contingency metrics across storms.	Same multi-storm design as Varuna's 8-storm test. Differs: the SAR classifier's own robustness, not terrain-index/DEM co-location failure.
Topographic index as prior for S-1 flood maps — 2023, Water (MDPI)	Adds HAND/topographic index as a Bayesian prior to S-1 maps; notes small gains on flat/urban terrain.	Supports Varuna's finding that terrain indices add little on flats. Differs: uses topography to <i>help mapping</i> ; Varuna uses a DEM ensemble to explain why topography can't decide street-scale ponding on flats.
Vertical accuracy of free global DEMs in flood-prone terrain — Marsh et al., 2024, Int. J. Digital Earth	Quantifies vertical error of global DEMs in floodplains (FABDEM best, RMSE ~1.24 m).	Empirical basis for Varuna's ±1 m FABDEM perturbation. Differs: <i>measures</i> DEM error; Varuna <i>propagates</i> it through the twin to measure co-location loss.
DEM-resolution effects on coastal flood vulnerability — Fereshtehpour & Karamouz, 2018, WRR	Monte-Carlo treatment of DEM error on flood extent; low-relief most sensitive.	Precedent for DEM Monte-Carlo ensembles. Differs: coastal bathtub; Varuna uses a spatially-correlated ensemble on a dynamic twin and reports a co-location fraction.
Modelling DEM errors in coastal flood inundation — Karamouz & Fereshtehpour, 2019, WRR	Models spatially non-stationary (autocorrelated) DEM error via stochastic realisations.	Justifies Varuna's <i>spatially-correlated</i> (not white-noise) perturbation. Differs: damage estimation, not SAR co-location diagnosis.
Elevation bias & topographic uncertainty in a lowland basin — 2025, J. Hydrology	Shows vertical error dominates flood extent/depth uncertainty in low-relief basins.	Confirms Varuna's core thesis (metre-scale noise > relief on flats). Differs: one-basin sensitivity mapping; Varuna ties it to SAR co-location across storms.
Evaluation of the NWM–HAND flood-mapping methodology — Johnson, Munasinghe, Eyelade, Cohen, 2019, NHESS	Evaluates the HAND terrain-index method vs 28 remotely-sensed maps.	HAND is one index Varuna tests. Differs: operational mapping product; Varuna uses HAND/TWI as <i>failing baselines</i> to isolate the DEM-noise mechanism.
Citizen observations contributing to flood modelling — Assumpção, Popescu, Jonoski, Solomatine, 2018, HESS	Reviews citizen/crowdsourced observations for flood monitoring/mapping/modelling; flags the lack of methods to use them in calibration/validation.	Foundational review Varuna's loop builds on. Differs: it identifies the gap ("no appropriate techniques"); Varuna supplies a concrete depth-band pin → reward-gated online update.
Can assimilation of crowdsourced data improve flood prediction? — Mazzoleni et al., 2017, HESS	Assimilates (synthetic) crowdsourced observations into hydrologic models.	Closest citizen-data→model-update prior. Differs: state assimilation with synthetic data and no deployment gate ; Varuna does reward-gated emulator fine-tuning on real reports with a ship/no-ship decision.
Improving hydrological models with crowdsourced data — Avellaneda et al., 2020, WRR	Assimilates <i>real</i> crowdsourced streamflow/stage into a hydrologic model.	Real-crowdsourced-data precedent. Differs: continuous filter update, no held-out promotion gate/rollback/audit log — Varuna's distinguishing safety layer.
Citizen science & crowdsourcing for pluvial flooding: a review — See et al., 2019, Frontiers in Earth Science	Reviews citizen-science/crowdsourcing for urban/pluvial flood applications.	Situates Varuna's participatory design. Differs: survey only; no online-learning system.
Crowdsourced social media for disaster mgmt: PetaJakarta — Holderness & Turpin, 2018, Computers, Environment & Urban Systems	Real-time Twitter-based flood crowd-mapping (PetaJakarta → PetaBencana) at megacity scale.	Best-known deployed citizen-flood system; direct comparator. Differs: <i>visualises</i> reports; does not feed a gated online-learning emulator or produce an auditable learning log.
NASA's Black Marble nighttime-lights product suite — Román et al., 2018, Remote Sensing of Environment	Defines the VNP46 daily VIIRS Day/Night-Band product (500 m), atmospherically/lunar-corrected.	Exact data product behind Varuna's outage overlay. Cited as method — Varuna operationalises it as a flood outage layer, not a remote-sensing advance.
Satellite assessment of electricity restoration after Hurricane Maria — Román et al., 2019, PLOS ONE	Uses Black Marble to track neighborhood-scale power outage/restoration post-disaster.	Canonical night-lights-for-outage study Varuna emulates. Differs: post-hoc recovery analysis; Varuna fuses the outage signal into a live flood dashboard.
Validation-gated multi-agent governance for online surrogate adaptation — Lim, Lee, Bang, 2026, arXiv	Champion-challenger promotion gate with shadow learning, safety-auditor & rollback for online surrogate updates.	Strongest cross-domain analog to Varuna's reward-gated loop (gate + rollback + audit). Differs: nuclear thermal-hydraulics, no citizen data, no flood/geospatial context — Varuna appears first to bring gated online updates + rollback + public audit log to a <i>flood</i> emulator driven by <i>citizen reports</i> .

2.5 Patents (all 21 URLs fetched & verified; summaries are not a legal claim-chart)

Patent	What it claims	Relation & key difference from Varuna
WO2025106413A1 — "Machine Learning Deluge" — Autodesk, 2024/25	Sequential CNN trained on ~10k stormwater sims produces flood maps 16–25× faster with interactive updates as a user places ponds/swales.	Closest patent to Varuna's fast ML twin. Differs: CNN surrogate of a CAD solver for design-time what-ifs — not a <i>differentiable satellite/DEM-built</i> twin; no street-graph GNN, live re-simulation-measured drainage, citizen online-learning, or aquifer coupling.
CN109902801A — Bayesian NN flood ensemble forecasting — Huazhong UST / Changjiang Water Res., 2019	Variational-inference Bayesian NN for flood forecasts with uncertainty.	Overlaps ML flood prediction. Differs: river-gauge time-series ensemble, not a spatial urban twin, no drainage graph or evacuation/recharge stack.
CN118692226A — AI-based flood analysis — Pearl River Water Res. Tech Center, 2024	Fuses history/terrain/weather/hydrology into an ML model refined by real-time monitoring to output risk maps/warnings.	Overlaps ML risk mapping. Differs: generic multi-source pipeline; no differentiable sim, street-graph GNN, or coupled drainage/recharge optimisation.
CN110991776A — GRU water-level prediction — Zhejiang Ponsnine, 2020	GRU predicts water level from rainfall + level over 6/12 h for early warning.	Overlaps ML forecasting. Differs: single-point RNN gauge forecast; no spatial twin, graph, or planning.
US12230116B2 — IoT smart-city flood early warning — Chengdu Qinchuan IoT, 2023/25	Fuses precipitation, road-ponding, GIS & drainage-capacity data across IoT to issue multi-level urban flood warnings.	Overlaps IoT urban warning + drainage awareness. Differs: rule/data-fusion warning platform; no learned simulator, self-supervised GNN, routing engine, or recharge.
US20130116920A1 — Flood-aware travel routing — IBM, 2011/13	Geo DB + route generator + flood simulator + risk model produce multiple routes scored by traveler flood risk.	Closest flood-safe-routing patent. Differs: offline-trained risk model feeding a GPS router; no differentiable city twin driving the risk surface, no online citizen updates, no drainage/recharge integration.
US20160047663A1 — Evacuation navigation device — Trimble, 2014/16	Handheld device dynamically re-routes occupants using maps + real-time environmental monitoring.	Overlaps dynamic evacuation. Differs: indoor/building egress, generic hazards, not flood-twin-driven city-scale routing.
US7349768B2 — Evacuation route planning tool — Boeing, 2005/08	GIS computes/recalculates area evacuation routes from road network, traffic & travel time.	Overlaps evacuation routing. Differs: general max-flow/min-cost planner; flood is one input, no learned flood sim, street-graph GNN, or recharge.
US9600997B1 — Localized flood alert system — IBM, 2015/17	UAV ultrasonic sensors measure local flood depth; personalized alerts + underwater-hazard map with individual safe paths.	Overlaps personalized alerts + safe routing. Differs: drone-sensor depth; no predictive twin, online-learning citizen reports, or drainage/recharge planning.
US20120264393A1 — Flood data collection & warning — IBM (now TSMC), 2011/12	Citizens report water level via SMS/call to "totem" nodes; system predicts level, gives feedback, auto-subscribes to alerts.	Closest crowdsourced/citizen-report patent. Differs: citizen input feeds a threshold predictor; Varuna's reward-gated online learning that updates the twin from validated reports is absent — no learned model-update loop.
US12094320B2 — Flood warning system — JM Torres & Assoc., 2022/24	Networked water-level sensor stations wirelessly trigger local visual warnings above thresholds.	Overlaps sensor early-warning. Differs: hardware threshold alarm; no modelling, ML, routing, or recharge.
US20170193305A1 — Flash-flooding detection — AGT International, 2015/17	Cameras + video analytics watch ground markers; loss of visibility signals flash-flood onset.	Overlaps detection/warning. Differs: computer-vision detection of present water; no forecasting twin, graph, or planning.
CN207924874U — Urban flood-control early-warning device — Guangdong Aike, 2018	Utility model fusing radar rainfall + in-drain sensors to forecast urban flooding & support public reporting.	Overlaps urban warning + drainage sensing + citizen input. Differs: hardware utility model; no learned simulator, GNN, routing, or recharge optimisation.
US20150134126A1 — Intelligent drainage system — IBM (now GlobalFoundries), 2013/15	Sensors feed a max-flow analysis that finds an optimal diversion path and actuates valves to prevent overflow.	Closest smart-drainage patent. Differs: real-time valve control on a <i>physical</i> network via max-flow; not a re-simulation-measured <i>street-graph drainage design</i> learned in a twin, and no recharge or evacuation layer.
WO2018215746A1 — Manhole and sewer network — Liverpool John Moores Univ., 2018	Dual-chamber manhole separating stormwater/sewage, boosting storage ~250 %.	Overlaps storm-drainage design/storage. Differs: physical hardware civil design, not a computational network-optimisation twin.
WO2020049310A1 — Smart sewer system — Environmental Monitoring Solutions, 2019/20	≥3 in-sewer level sensors detect blockages from anomalous level patterns to pre-empt flooding.	Overlaps smart-drainage monitoring. Differs: blockage detection only; no design optimisation, twin, or planning.
US9638334B2 — Flow-control gate for detention pond — 3 Rocks Engineering, 2015/17	Passive float-on-track gate holds detention-pond discharge at pre-development rates without power.	Overlaps detention/retention control. Differs: passive mechanical outlet; no siting logic, predictive control, or twin.
WO2009138995A1 — Rainwater harvesting/recharge trench — V.K. Kedia (India), 2008/09	Underground PVC-lined boundary trench captures ~80 % rainfall as artificial recharge, cutting runoff/erosion.	Overlaps stormwater-to-groundwater recharge (Indian applicant). Differs: passive civil structure; no metered recharge control or flood-to-aquifer coupling model/optimisation.
US9011689B1 — Artificial recharge system — US Dept. of the Interior, 2012/15	Constructed wetland + settling pond + recharge gallery recharges aquifers without chemicals.	Overlaps MAR. Differs: physical treatment-and-recharge apparatus; no control model coupling flood runoff to metered recharge, no digital twin.
US7721799B2 — Flow-control packer & ASR system — 2006/10	Downhole packer regulates injection/recovery for ASR wells.	Overlaps groundwater recharge control hardware. Differs: well-bore flow hardware; no surface stormwater-to-aquifer optimisation or twin.
WO2022118101A1 — Remote dam monitoring — Rezatec, 2021/22	Geospatial AI on satellite SAR (PS-InSAR) + optical detects ground motion/seepage on dams.	Closest SAR/satellite-system patent. Differs: dam/embankment deformation & seepage monitoring, not SAR-derived flood-extent <i>building of a differentiable urban twin</i> .

Patent white-space found: no located patent (a) builds a **differentiable / simulation-ready city flood twin from satellite+DEM data**, (b) uses a **self-supervised street-graph GNN** for urban flood risk (only academic work exists — FloodGNN-GRU, physics-informed GNNs), or (c) couples **flood runoff to metered aquifer recharge as a twin-optimised objective** (the recharge patents are all *passive/physical* infrastructure). These three axes are the strongest patent-novelty highlights.

3. Head-to-head novelty matrix (the bottom line)

For each Varuna component: the **single closest prior art**, the **precise differentiator**, and a **verdict**. Verdicts: **Novel combination** (each ingredient exists, the assembly does not) · **Novel** (no located equivalent) · **Differentiated** (close prior art, clear distinguishing feature) · **Not claimed novel** (established method used as-is).

Varuna component	Closest prior art	How Varuna differs	Verdict
V1 Free-tier differentiable satellite→twin→dashboard	Inunda (2026); Liu 2025 WRR; DassFlow2D	Adds a U-Net emulator + free global open-data ingest (FABDEM/WorldCover/JRC/S-1) + hosted public dashboard on free-tier compute for data-scarce cities; Inunda is a US-data pure solver, near-concurrent	Differentiated (core engine = Inunda; the <i>deployed open free-tier system</i> is the delta)
V2 Honest SAR validation + DEM-ensemble co-location diagnosis	Bernhofen 2018; Wing 2017; Marsh 2024; Fereshtehpour 2018	Turns "low CSI on flats" into a quantified mechanism — a spatially-correlated ±1 m DEM ensemble giving a co-location fraction (~5 % flat vs ~52 % hilly); prior work benchmarks or measures DEM error but does not couple it to multi-storm SAR co-location as an honest-negative	Novel (diagnosis); metrics/CSI not claimed novel
V3 Street-graph spiderweb drainage, benefit re-simulated, flood-safe non-river outfalls	Hesarkazzazi 2022 (street-graph layout MOO); Afshar 2005 (GA+SWMM); IBM US20150134126A1	Re-simulates the 2-D twin to measure the cut (not optimise pipe cost against a hydraulic model); reverse multi-source Dijkstra from explicitly flood-safe non-river outfalls ; density-as-lever finding	Novel combination (re-sim-measured street-graph routing with flood-safe outfalls not found)
V4 FloodGNN: self-supervised, FiLM-rainfall, street-graph GNN + routing + zero-shot transfer	SWE-GNN/mSWE-GNN (Bentivoglio); FloodGNN-GRU & ICML (Kazadi); ASTGCN (Farahmand)	Combines 5 ingredients none co-occupy : OSM <i>street-graph</i> nodes · FiLM continuous-rainfall single model · self-supervision purely on twin labels · evacuation routing · zero-shot flat↔hilly transfer . Closest works use mesh nodes, autoregressive rollout, single-storm training, or observed-flood supervision	Novel combination
V5 Buildability-constrained, phased, re-measured detention siting	LID optimisation (SWMMLIDopt; J. Hydrology 2018; SWMM-FTC)	Excludes building footprints & permanent water, allows under-street , sizes each site to its <i>local pooled depth</i> , re-measures the cut, emits a 10/20/40 % phased costed ladder ; LID work optimises green controls at parcel scale without buildability/pooled-depth logic	Differentiated
V6 Flood-to-drought / managed-aquifer-recharge coupling	UTFI/IWMI 2015 (rural); MAR-urban 2026; Guadalajara 2022; Atal Bhujal Yojana	Couples urban flood-cut with aquifer recharge on the city street graph with a Recharge Suitability Index + metered recharge plans, framed to Indian norms; closest works are rural-basin (UTFI) or conceptual/site-suitability urban MAR without engineered drainage design + measured flood cut	Novel combination (urban, engineered, measured coupling)
V7 Reward-gated, rollback-protected, publicly-audited citizen online-learning loop	Mazzoleni 2017 / Avellaneda 2020 (crowdsourced assimilation); PetaBencana; Lim 2026 (nuclear, cross-domain)	Ship/no-ship gate (held-out-by-day improvement + all-wet simulation-replay anchor + bootstrap win-rate) with a public audit log ; flood crowdsourcing prior does continuous assimilation or visualisation, no gated update — nearest gated-update analog is outside geoscience	Novel (first-of-kind for a flood emulator on citizen reports)
V8 Seeded-collapse reproducibility hazard + detect-and-reseed guard	General ML reproducibility literature (no direct flood analog located)	Documents that fixed seeds make failures reproducible too — a deterministic softplus-head collapse with a <i>frozen-metric tell</i> — and a detect-and-reseed guard; no located flood/emulator paper reports this failure mode	Novel (as a documented finding)
V9 Grounded multilingual advisories + VIIRS outage overlay	Román 2018/2019 (Black Marble); LLM-grounding practice	Integration/application: hallucination-guarded local-language briefs from <i>only</i> system numbers + a night-lights outage layer with a "unknown ≠ dark" cloud-mask caveat, inside one flood dashboard	Not claimed novel (established methods integrated)

4. Overall assessment

No single system or patent was found that combines Varuna's stack — a differentiable Bates-2010 PyTorch twin + U-Net emulator + free global open satellite data + self-supervised street-graph GNN with routing & transfer + re-simulation-measured street-graph drainage + buildability-constrained detention + flood-to-aquifer-recharge coupling + reward-gated citizen online learning — served free-tier for data-scarce Indian cities. Each *ingredient* has strong prior art (catalogued above); the **assembly, and four specific mechanisms, are where novelty concentrates**:

1. **DEM-ensemble co-location diagnosis (V2)** — a genuinely novel, honest-negative contribution.
2. **Re-simulation-measured street-graph drainage with flood-safe non-river outfalls (V3)** — a design + evaluation loop not found in the drainage-optimisation literature or patents.
3. **FloodGNN's five-ingredient combination (V4)** — no located work co-occupies street-graph + FiLM-rainfall + twin-self-supervision + routing + zero-shot transfer.
4. **Reward-gated, audited citizen online-learning for a flood emulator (V7)** — first-of-kind; nearest analog is in nuclear surrogate modelling.

The one thing to foreground honestly: *Inunda* (*arXiv*, July 2026) is a near-concurrent, near-identical differentiable Bates-2010-in-PyTorch core. It must be cited as **related work you distinguish** (emulator + open-data + hosted free-tier twin + the whole planning/GNN/recharge/citizen stack), **not** as something Varuna predates. Do **not** claim novelty on: the Bates-2010 scheme, FiLM, HAND/TWI indices, VIIRS Black Marble, or ASR/MAR as concepts — cite them as methods/baselines.

5. Suggested targeted follow-ups (gaps this search could not fully close)

- A dedicated search for any tool literally doing **OSM-street-graph drainage routing with hydraulic re-simulation of the benefit** (street-graph *layout* optimisation exists; the re-sim-measured loop appears to be a real gap — worth confirming).
 - **Indian urban UTFI extensions** (rural UTFI is well-covered; urban is only just emerging — e.g. a 2024 "rural–urban approach" paper).
 - A **legal claim-chart** (not just abstract reading) on the four closest patents — WO2025106413A1 (Autodesk), US20150134126A1 & US20130116920A1 & US20120264393A1 (IBM) — before any IP filing.
 - Full-text verification of the paywalled items flagged in §2 (Liu 2025 WRR; a few ScienceDirect/Springer/Nature/Wiley DOIs auto-fetch-blocked but browser-resolvable) before formal submission.
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6. What to cite where, in one line

- **Differentiable twin (V1):** cite Bates 2010 (method), LISFLOOD-FP 8.0 (baseline), *Inunda* 2026 & Liu 2025 WRR (concurrent differentiable), position as *open free-tier deployed system*.
 - **Validation (V2):** cite Wing 2017 & Bernhofen 2018 (CSI validation), Marsh 2024 & Fereshtehpour 2018 (DEM error), claim the *co-location ensemble diagnosis*.
 - **Drainage (V3) & storage (V5):** cite Hesarkazzazi 2022 & Afshar 2005 & the LID-optimisation set, claim *re-sim-measured street-graph routing + buildability + flood-safe outfalls*.
 - **FloodGNN (V4):** cite SWE-GNN/mSWE-GNN, FloodGNN-GRU, ASTGCN, Perez 2018 (FiLM), Kratzert 2019 (transfer), claim the *five-ingredient combination*.
 - **Recharge (V6):** cite UTFI/IWMI 2015, MAR-urban 2026, Guadalajara 2022, Atal Bhujal Yojana, claim *urban engineered measured flood↔recharge coupling*.
 - **Citizen loop (V7):** cite Assumpção 2018, Mazzoleni 2017, Avellaneda 2020, PetaBencana, Lim 2026 (cross-domain), claim the *gated audited update*.
 - **Outage/advisories (V9):** cite Román 2018/2019 (Black Marble) as method.
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7. Sources & links (full URLs)

Grouped in the order they appear in §2. Every URL below was returned by search on 2026-07-23; load-bearing items were fetched. Paywalled/auto-fetch-blocked pages are canonical DOIs that resolve in a browser.

Flood digital twins & differentiable models

1. Inunda (Zhi Li, 2026) — <https://arxiv.org/html/2607.09614>
2. Universal SWE / Differentiable Programming (Liu et al., 2025, WRR) — <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025WR040265>
3. Bates, Horritt & Fewtrell 2010 (J. Hydrology) — <https://doi.org/10.1016/j.jhydrol.2010.03.027>
4. LISFLOOD-FP 8.0 (Shaw et al., 2021, GMD) — <https://gmd.copernicus.org/articles/14/3577/2021/>
5. Subgrid local-inertial urban flood (Nithila Devi & Kuiry, 2024, WRR) — <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2023WR035334>
6. SFINCS (Deltares) — <https://github.com/Deltares/SFINCS>
7. Fathom Global Flood Map — <https://www.fathom.global/product/global-flood-map/>
8. JBA Global Flood Models — <https://jbagr.com/digital-tools/global-flood-models/>
9. Previsico Flood Intel Platform — <https://previsico.com/en-us/intel-platform>
10. FloodCast / GeoPINS (Xu et al., 2024) — <https://arxiv.org/abs/2403.12226> · code <https://github.com/HydroPML/FloodCast>
11. Deep CNN rapid fluvial inundation (Kabir et al., 2020) — <https://arxiv.org/pdf/2006.11555>
12. DassFlow2D-V3 (Pujol et al., 2022, GMD) — <https://gmd.copernicus.org/articles/15/6085/2022/>
13. Adjoint assimilation for 2-D urban flood (Zhang et al., 2024) — <https://www.sciencedirect.com/science/article/abs/pii/S0022169424012812>
14. smash v1.0 (Colleoni et al., 2025, GMD) — <https://gmd.copernicus.org/articles/18/7003/2025/>
15. Fully differentiable rainfall-runoff (2025, HESS) — <https://hess.copernicus.org/articles/29/6257/2025/>
16. CLDNet AI digital twin (2026) — <https://arxiv.org/html/2605.13761v1>
17. FlowsDT geospatial digital twin (2026) — <https://www.sciencedirect.com/science/article/pii/S0198971526000165>
18. Digital Twins for Urban Flood Risk Mgmt review (2025, Remote Sensing) — <https://www.mdpi.com/2072-4292/17/17/3104>
19. Probabilistic flood-risk mapping Indian megacities / ALERT (2025) — <https://www.sciencedirect.com/science/article/pii/S2590123025039246>
20. Copernicus/Sentinel-1 GFM (2025, RSE) — <https://www.sciencedirect.com/science/article/pii/S0034425725005127>

AI / GNN flood prediction & routing

21. Google Flood Hub — global extreme floods (Nearing et al., 2024, Nature) — <https://www.nature.com/articles/s41586-024-07145-1> · OA <https://pmc.ncbi.nlm.nih.gov/articles/PMC10954541/>
22. Predictions in ungauged basins with ML (Kratzert et al., 2019, WRR) — <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2019WR026065>
23. SWE-GNN (Bentivoglio et al., 2023, HESS) — <https://hess.copernicus.org/articles/27/4227/2023/>
24. mSWE-GNN (Bentivoglio et al., 2025, NHSS) — <https://nhess.copernicus.org/articles/25/335/2025/>
25. FloodGNN-GRU (Kazadi et al., 2024, Env. Data Science) — <https://www.cambridge.org/core/journals/environmental-data-science/article/floodgnngru-a-spatiotemporal-graph-neural-network-for-flood-prediction/93BA1DA8D6ECC93D985656C3BC1EA3DE>
26. Pluvial Flood Emulation w/ Hydraulics-informed MP (Kazadi et al., 2024, ICML) — <https://proceedings.mlr.press/v235/kazadi24a.html>
27. ASTGCN urban flood nowcasting (Farahmand et al., 2023, Sci. Reports) — <https://www.nature.com/articles/s41598-023-32548-x> · preprint <https://arxiv.org/abs/2111.08450>
28. Interpretable physics-informed GNNs (Taghizadeh et al., 2025, CACIE) — <https://onlinelibrary.wiley.com/doi/full/10.1111/mice.13484>
29. GNN surrogate for urban drainage hydraulics (Zhang et al., 2024, Water Research) — <https://www.sciencedirect.com/science/article/abs/pii/S0043135424010418> · preprint <https://arxiv.org/abs/2404.10324>
30. GNN route recommendation under hazards (Liu & Meidani, 2025) — <https://arxiv.org/abs/2501.09803>
31. Flood model + Dijkstra risk-avoidance routing (2023, WRM) — <https://link.springer.com/article/10.1007/s11269-023-03500-5>
32. Dam-failure evacuation routing (2025, Applied Sciences) — <https://www.mdpi.com/2076-3417/15/8/4518>
33. Transferability of urban pluvial flood-depth models, Berlin (Seleem et al., 2023, NHSS) — <https://nhess.copernicus.org/articles/23/809/2023/>
34. Cross-region streamflow ED-DLSTM (Li et al., 2024, The Innovation) — [https://www.cell.com/the-innovation/fulltext/S2666-6758\(24\)00055-9](https://www.cell.com/the-innovation/fulltext/S2666-6758(24)00055-9)

35. Data-driven flood emulation deep CNNs (Guo et al., 2021) — <https://arxiv.org/abs/2004.08340>
36. U-RNN urban flood nowcasting (Jiang et al., 2025, J. Hydrology) — <https://www.sciencedirect.com/science/article/abs/pii/S002216942500455X>
37. Multi-fidelity GNNs for flood-hazard mapping (2025, RESS) — <https://www.sciencedirect.com/science/article/abs/pii/S136481522500338X>
38. Deep learning for flood mapping: review (Bentivoglio et al., 2022, HESS) — <https://hess.copernicus.org/articles/26/4345/2022/>
39. FiLM (Perez et al., 2018, AAAI) — <https://arxiv.org/abs/1709.07871>

Drainage optimisation & managed aquifer recharge

40. Graph-theory MOO drainage layouts (Hesarkazzazi et al., 2022) — <https://www.sciencedirect.com/science/article/pii/S2210670722001548>
41. Structural optimisation of urban drainage (2024, Water) — <https://www.mdpi.com/2073-4441/16/12/1696>
42. Network structure optimisation w/ redundancies (2022, IJDRS) — <https://link.springer.com/article/10.1007/s13753-022-00445-y>
43. Hydraulic-based drainage-network optimisation (2024, Applied Water Science) — <https://link.springer.com/article/10.1007/s13201-024-02204-4>
44. Hydrograph-based storm-sewer GA (Afshar et al., 2005, CJCE) — <https://cdns.copernicus.com/doi/10.1139/I05-121>
45. MOO of LID controls (2018, J. Hydrology) — <https://www.sciencedirect.com/science/article/abs/pii/S0022169418303214>
46. SWMMLIDopt (2024, J. Hydroinformatics) — <https://iwaponline.com/jh/article/26/11/2781/105364/>
47. Spatial optimisation LID SWMM-FTC (2022, JEM) — <https://www.sciencedirect.com/science/article/abs/pii/S0301479722001475>
48. Sponge City review China (2023, Water Sci. & Tech.) — <https://iwaponline.com/wst/article/88/10/2499/98252/>
49. Sponge City Shenzhen evidence (2025, Urban Science) — <https://doi.org/10.3390/urbansci10030135>
50. EAR Using Stormwater: State of Science (US EPA, 2021) — <https://www.gwpc.org/wp-content/uploads/2022/08/EAR-Using-Stormwater-State-of-Science-Review-USEPA2021.pdf>
51. MAR for urban floods + groundwater depletion (2026, STOTEN) — <https://www.sciencedirect.com/science/article/abs/pii/S0048969726001580>
52. Lessons Learned from ASR in the US (GWPC, 2022) — <https://www.gwpc.org/wp-content/uploads/2022/08/Lessons-Learned-from-Aquifer-Storage-and-Recovery-ASR-Systems-in-the-United-States.pdf>
53. FEMA Innovative Drought & Flood Mitigation (2017) — https://www.fema.gov/sites/default/files/documents/fema_innovative-drought-flood-mitigation-projects.pdf
54. Atal Bhujal Yojana (CGWB) — <https://www.cgwb.gov.in/en/participatory-ground-water-management-atal-bhujal-yojana>
55. Legislation on Rainwater Harvesting, Chennai/TN (CSE India) — <https://www.cseindia.org/legislation-on-rainwater-harvesting-1111>
56. UTFI Ganges (Pavelic et al. / IWMI, 2015) — <https://cgspace.cgiar.org/items/7d13ba30-dba2-42c0-af1b-f596b044cd64>
57. Floodwater recharge offsetting depletion, Ramganga (2017, SWRM) — <https://link.springer.com/article/10.1007/s40899-017-0168-6>
58. Managing underground transfer of floods, Ramganga (2019, J. Hydrology) — <https://www.sciencedirect.com/science/article/abs/pii/S0022169419312533>
59. Urban flood mitigation via deep aquifer recharge, Guadalajara (2022, IJERPH) — <https://pmc.ncbi.nlm.nih.gov/articles/PMC8949559/>

SAR validation, DEM uncertainty, citizen science & night-lights

60. Validation of 30 m flood model CONUS (Wing et al., 2017, WRR) — <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1002/2017WR020917>
61. Collective validation of global flood models (Bernhofen et al., 2018, ERL) — <https://iopscience.iop.org/article/10.1088/1748-9326/aae014>
62. Sen1Floods11 (Bonafilia et al., 2020, CVPR-W) — <https://ieeexplore.ieee.org/document/9150760/>
63. Robustness of Bayesian S-1 flood mapping (2025, Sci. Remote Sensing) — <https://www.sciencedirect.com/science/article/pii/S2666017225000161>
64. Topographic index prior for S-1 flood maps (2023, Water) — <https://www.mdpi.com/2073-4441/15/23/4034>
65. Vertical accuracy of global DEMs in floodplains (Marsh et al., 2024, IJDE) — <https://www.tandfonline.com/doi/full/10.1080/17538947.2024.2308734>
66. DEM-resolution effects on coastal flood vulnerability (Fereshtehpour & Karamouz, 2018, WRR) — <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2017WR022318>
67. Modelling DEM errors in coastal inundation (Karamouz & Fereshtehpour, 2019, WRR) — <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2018WR024562>

68. Elevation bias & topographic uncertainty, lowland basin (2025, J. Hydrology) — <https://www.sciencedirect.com/science/article/abs/pii/S0022169425021730>
69. Evaluation of NWM–HAND flood mapping (Johnson et al., 2019, NHES) — <https://nhess.copernicus.org/articles/19/2405/2019/>
70. Citizen observations in flood modelling (Assumpção et al., 2018, HESS) — <https://hess.copernicus.org/articles/22/1473/2018/>
71. Assimilation of crowdsourced data for flood prediction (Mazzoleni et al., 2017, HESS) — <https://hess.copernicus.org/articles/21/839/2017/>
72. Improving hydrological models w/ crowdsourced data (Avellaneda et al., 2020, WRR) — <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2019WR026325>
73. Citizen science & crowdsourcing for pluvial flooding review (See et al., 2019, Frontiers) — <https://www.frontiersin.org/journals/earth-science/articles/10.3389/feart.2019.00044/full>
74. PetaJakarta / PetaBencana (Holderness & Turpin, 2018, CEUS) — <https://www.sciencedirect.com/science/article/abs/pii/S0198971518301066>
75. NASA Black Marble product suite (Román et al., 2018, RSE) — <https://www.sciencedirect.com/science/article/pii/S003442571830110X>
76. Electricity restoration after Hurricane Maria (Román et al., 2019, PLOS ONE) — <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0218883>
77. Validation-gated online surrogate adaptation (Lim et al., 2026) — <https://arxiv.org/abs/2606.03321>

Patents

78. WO2025106413A1 Machine Learning Deluge (Autodesk) — <https://patents.google.com/patent/WO2025106413A1/en>
79. CN109902801A Bayesian NN flood forecasting — <https://patents.google.com/patent/CN109902801A/en>
80. CN118692226A AI flood analysis — <https://patents.google.com/patent/CN118692226A/en>
81. CN110991776A GRU water-level prediction — <https://patents.google.com/patent/CN110991776A/en>
82. US12230116B2 IoT smart-city flood warning — <https://patents.google.com/patent/US12230116B2/en>
83. US20130116920A1 Flood-aware travel routing (IBM) — <https://patents.google.com/patent/US20130116920A1/en>
84. US20160047663A1 Evacuation navigation device (Trimble) — <https://patents.google.com/patent/US20160047663A1/en>
85. US7349768B2 Evacuation route planning (Boeing) — <https://patents.google.com/patent/US7349768B2/en>
86. US9600997B1 Localized flood alert (IBM) — <https://patents.google.com/patent/US9600997B1/en>
87. US20120264393A1 Flood data collection & warning (IBM) — <https://patents.google.com/patent/US20120264393A1/en>
88. US12094320B2 Flood warning system — <https://patents.google.com/patent/US12094320B2/en>
89. US20170193305A1 Flash-flooding detection (AGT) — <https://patents.google.com/patent/US20170193305A1/en>
90. CN207924874U Urban flood-control warning device — <https://patents.google.com/patent/CN207924874U/en>
91. US20150134126A1 Intelligent drainage system (IBM) — <https://patents.google.com/patent/US20150134126A1/en>
92. WO2018215746A1 Manhole and sewer network (LJMU) — <https://patents.google.com/patent/WO2018215746A1/en>
93. WO2020049310A1 Smart sewer system — <https://patents.google.com/patent/WO2020049310A1/en>
94. US9638334B2 Flow-control gate for detention pond — <https://patents.google.com/patent/US9638334B2/en>
95. WO2009138995A1 Rainwater harvesting/recharge trench (India) — <https://patents.google.com/patent/WO2009138995A1/en>
96. US9011689B1 Artificial recharge system (US DOI) — <https://patents.google.com/patent/US9011689B1/en>
97. US7721799B2 Flow-control packer & ASR — <https://patents.google.com/patent/US7721799B2/en>
98. WO2022118101A1 Remote dam monitoring (Rezatec) — <https://patents.google.com/patent/WO2022118101A1/en>
99. (SAR flood-index, academic) CN113361403A NDFI SAR flood extraction — <https://patents.google.com/patent/CN113361403A/en>